

Hands-On Labs - Create Short Video Film using AI

WSQ Course Code: TGS-2020505925 | Version v12.0 | 27 September 2026 Conducted by: Tertiary Infotech Academy Pte Ltd (UEN 201200696W)

Twelve labs across the five delivery topics, all building one 48-second short film, *The Last Ferry* - from the need analysis and the reference sheets to the final export ladder. Every folder is independently usable: it carries its own steps, data, local media and prompt pack, and needs no network access once the Python dependencies are installed.

Lab	Title	Topic	Outcome	A	K	In-class
01	Film Needs Analysis and AI Production Pipeline Specification	01	ELO1	A1	K1, K2	within LU1 facilitation
02	Frames, Formats and Aspect Ratios for Film Delivery	02	ELO2	A2	K3	15 min
03	Structured Prompting for Cinematic Stills	02	ELO2	A2	K3, K4	15 min
04	State Variants, Clean-Up and Reframing with Masks	02	ELO2	A2	K4	15 min
05	Cinematic Grading and Enhancement Measured with PSNR and SSIM	02	ELO2	A2	K4	15 min
06	Character Reference Sheets and Consistency Features	03	ELO3	A4	K5, K6	30 min
07	Look Bible and Scene Continuity with Global Descriptors	03	ELO3	A3	K7	30 min
08	Cast-Presence Detection and Detector Evaluation	04	ELO4	A5	K8, K9	60 min

Lab	Title	Topic	Outcome	A	K	In-class
09	Storyboard to Shot Prompts, Start and End Frames, and the Animatic	04	ELO4	A5, A6	K8, K10	30 min
10	Shot, Motion and Continuity Analytics on the Assembled Sequence	04	ELO5	A6	K10	60 min
11	Voiceover, Music, Captions and the Final Export Ladder	04	ELO4	A5, A6	K9, K10	30 min
12	Cloud, Local and Hybrid Production Architecture Evaluation	05	ELO6	A7, A8	K11, K12, K13	60 min

Total scheduled practical time: 360 minutes (6 hours) - matching the 360-minute practical allocation in the approved course proposal.

Coverage

Code	Statement	Labs
A1	Identify the needs of vision systems technology in industrial applications	01
A2	Apply the principles of processing, filtering and analysis methods for video data	02, 03, 04, 05
A3	Analyse global feature descriptions	07
A4	Design and implement feature extraction and representation methods	06
A5	Design and apply machine-learning based methods for object detection, object tracking and activity recognition	08, 09, 11
A6	Design and apply video analytics algorithms for high-level video analytics tasks	09, 10, 11
A7	Design the architecture of applied vision systems	12
A8	Design, develop and evaluate edge-based and cloud-based systems	12
K1	Vision system concepts	01
K2	Business applications of vision systems	01
K3	Methods to represent image and video data	02, 03
K4	Image and video processing, filtering and transformation methods	03, 04, 05
K5	Feature extraction and representation techniques	06
K6	Local feature descriptions, edge, colour, texture and motion	06
K7	Global feature descriptions, statistical and geometrical methods	07
K8	Deep learning concepts	08, 09
K9	Object segmentation, detection and recognition	08, 11
K10	Activity tracking, generative models, scene understanding and event discovery	09, 10, 11
K11	Vision system architecture	12
K12	Vision communication protocols	12
K13	Real-world design constraints and solution options	12

Environment

```
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\Activate.ps1
pip install --upgrade pip
pip install "opencv-python>=4.10" numpy
```

No lab requires a paid service. Where a generative model can optionally be used - for stills, clips, voice or music - the lab supplies a local reference set so the deliverable can be completed offline, and asks you to record which path you took. FFmpeg is optional (Lab 11 uses it to mux audio when present).

Provenance rules that apply to every lab

1. A SIMULATED asset is a deterministic OpenCV illustration. It is never described as a photograph and never as generative-model output.
2. An AI-generated reference is copied unchanged and travels with a PROVENANCE.md recording the tool, date and prompt.
3. The label travels to every derivative: a crop or reframe of a simulated clip is re-labelled on export.
4. An ANIMATIC is labelled on every frame and is never submitted as a generated film.
5. Anything you generate yourself is recorded with the model ID, the prompt version and the watermark expectation.

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